

iConnect Plus — Master Architecture Document

Document type: Solution Architecture & Shipping Master
Product: iConnect Plus (Enterprise ERP)
Baseline: Architecture Lock v1.1 · Addendum v1.2 · ADR-001 · ADR-002
Updated: 2026-10-09
Classification: Internal — Confidential
Audience: Engineering, Architecture, DevOps, Product

This is the **master architecture document** for **iConnect Plus** as implemented and as locked. It consolidates system design, the locked technology stack, platform seams, module features and flows, Mermaid architecture diagrams, and shipping strategies for **Coolify/AWS**, **Compose**, and **on-premise**. Calendar timelines and month-based phase dates are intentionally omitted.

For day-to-day module detail prefer `docs/00_CURRENT/`. This document is the consolidated architecture view aligned with Addendum v1.2.

Contents

#	Section
1	Executive Summary
2	Product Scope and Module Map
3	Architectural Principles
4	High-Level System Architecture
5	Technology Stack (Current / Locked)
6	Clean Architecture and Request Flow
7	Multi-Tenancy and Security
8	Data and Storage Architecture
9	Platform Engines
10	Cross-Domain Business Flows
11	Module Catalog — Full Portfolio
12	Roadmap Platform Extensions
13	Integration Strategy
14	On-Premise Deployment Plan
15	AWS Deployment Plan
16	DevOps, Environments, and CI/CD
17	Quality, Observability, and Compliance
18	Key Risks and Mitigations
19	Document Hierarchy and Compliance

1. Executive Summary

iConnect Plus is a **modular monolith** ERP covering foundation platform services and twenty-plus business domains — Finance, CRM, Sales, Procurement, Inventory, Manufacturing, Quality, HR, Payroll, Projects, Assets, Service, Helpdesk, Documents, GRC, Analytics, Integration, Ecommerce, Portal, Marketing, and employee self-service — plus platform AI surfaces (`agent_read` , `voice_agent` , MCP) and roadmap extensions for Virtual E.A., Licensing, and Backup & DR.

Every module shares:

- One identity and RBAC model (permission catalogs + `HttpOnly` JWT cookies)
- One design system (Next.js + Tailwind + ShadCN · UI/UX Pro Max)
- One API surface (`/api/v1`)
- One transactional database (PostgreSQL) with tenant isolation
- Shared Workflow, Notification, Audit, and Integration engines
- Shared `modules.platform` seams (ports, outbox, numbering, approvals, encryption, SLOs)

Central decisions (locked + current shipping):

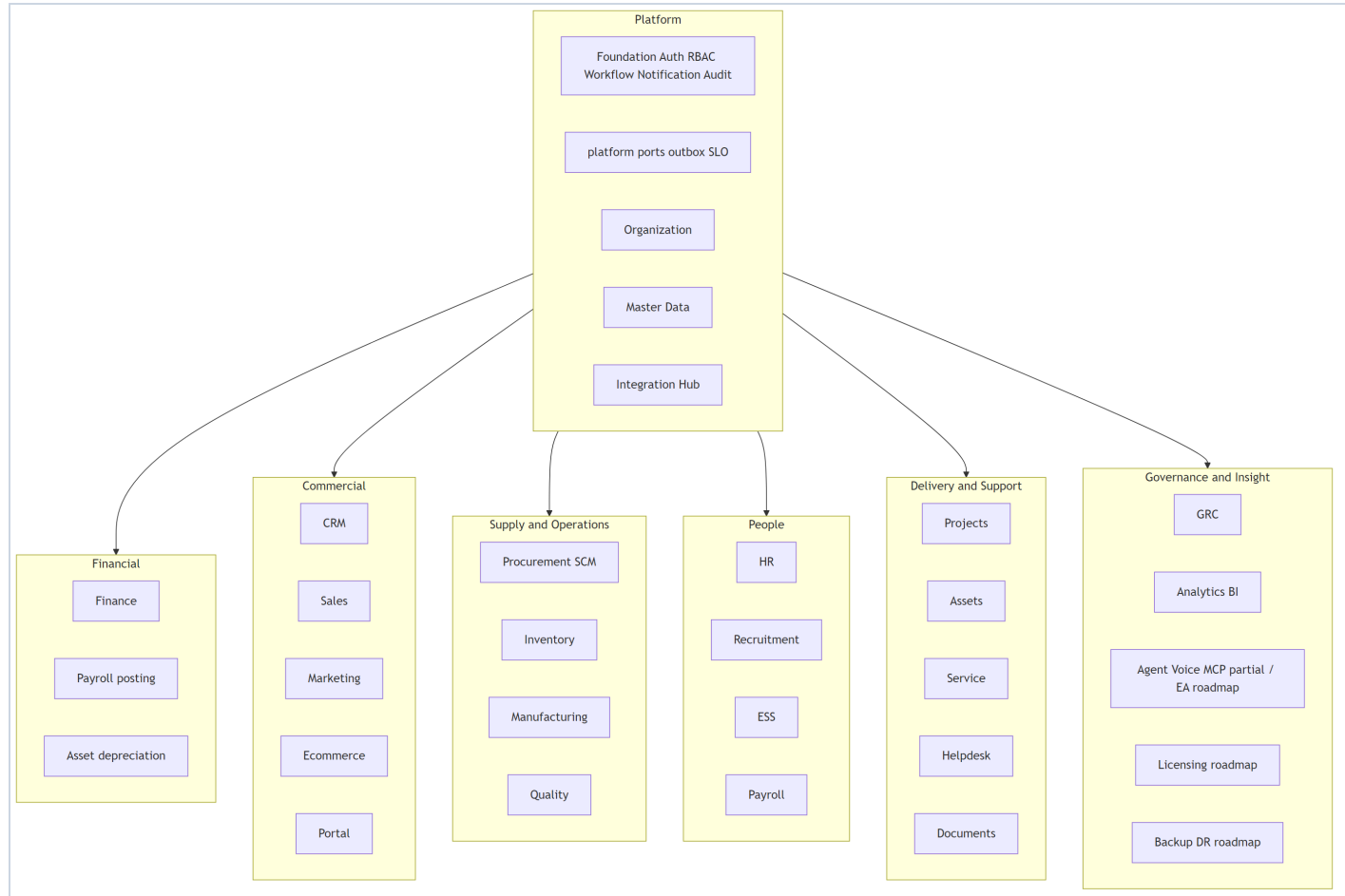
Decision	Choice
Architecture	Modular monolith · Clean Architecture · DDD (ADR-001)
Backend	Python 3.13+ · FastAPI · SQLAlchemy 2 · Alembic · Pydantic v2 · Celery (ADR-002)
Frontend	Next.js 16+ · TypeScript · Tailwind · ShadCN · Zod
OLTP	PostgreSQL
Cache / broker	Redis · RabbitMQ
Objects	MinIO (local/on-prem) · AWS S3 (cloud)
Search	OpenSearch — evolution-gated (not required for OLTP)
Auth	Microsoft Entra SSO · JWT in HttpOnly cookies
Deploy artifact	Docker images (API, worker, beat, web, employee-app)
SaaS shipping today	Coolify on EC2 · RDS PostgreSQL · S3 · Compose (<code>docker-compose.coolify.yml</code>)
Cloud target (gated)	EKS/ECS · ElastiCache · Amazon MQ · OpenSearch when Tier D readiness is met
On-prem	Same containers · Compose / K3s / customer Kubernetes

SaaS and on-premise share **one codebase and one image pipeline**; configuration and infrastructure topology differ by destination.

2. Product Scope and Module Map

2.1 Domain grouping

Figure 1



2.2 Module portfolio (summary)

Domain	Modules	Shared core entities
Platform	Foundation, Email/Notifications, Organization, Master Data, Integration Hub, platform seams	Tenant, user, role, company, branch, employee/customer/vendor/product masters, outbox
Financial	Finance, Payroll (GL side), Assets (depreciation)	COA, journal, period, cost center, tax, currency
Commercial	CRM, Sales, Marketing, Ecommerce, Portal	Account, lead, opportunity, order, invoice, campaign
Supply	Procurement, Inventory, Manufacturing, Quality, SCM (ops)	Item, vendor, UoM, BOM, WO, inspection, GRN
People	HR, Recruitment, ESS, Payroll	Employee, attendance, leave, salary structure
Delivery	Projects, Assets, Service, Helpdesk, DMS	Project, asset, ticket, document, SLA
Insight	Analytics, GRC, Agent/Voice/MCP (partial), Licensing (roadmap), Backup/DR (roadmap)	KPI, policy, risk, entitlement, backup job

2.3 Sell-together and sell-alone

Every module must satisfy:

1. **Licensing contract** — entitlement service (roadmap) disables unlicensed APIs, UI, and jobs (not merely hides menus).
2. **Degradation contract** — defined behavior when a sibling module is absent (e.g. Sales exports invoices when Finance is not licensed).

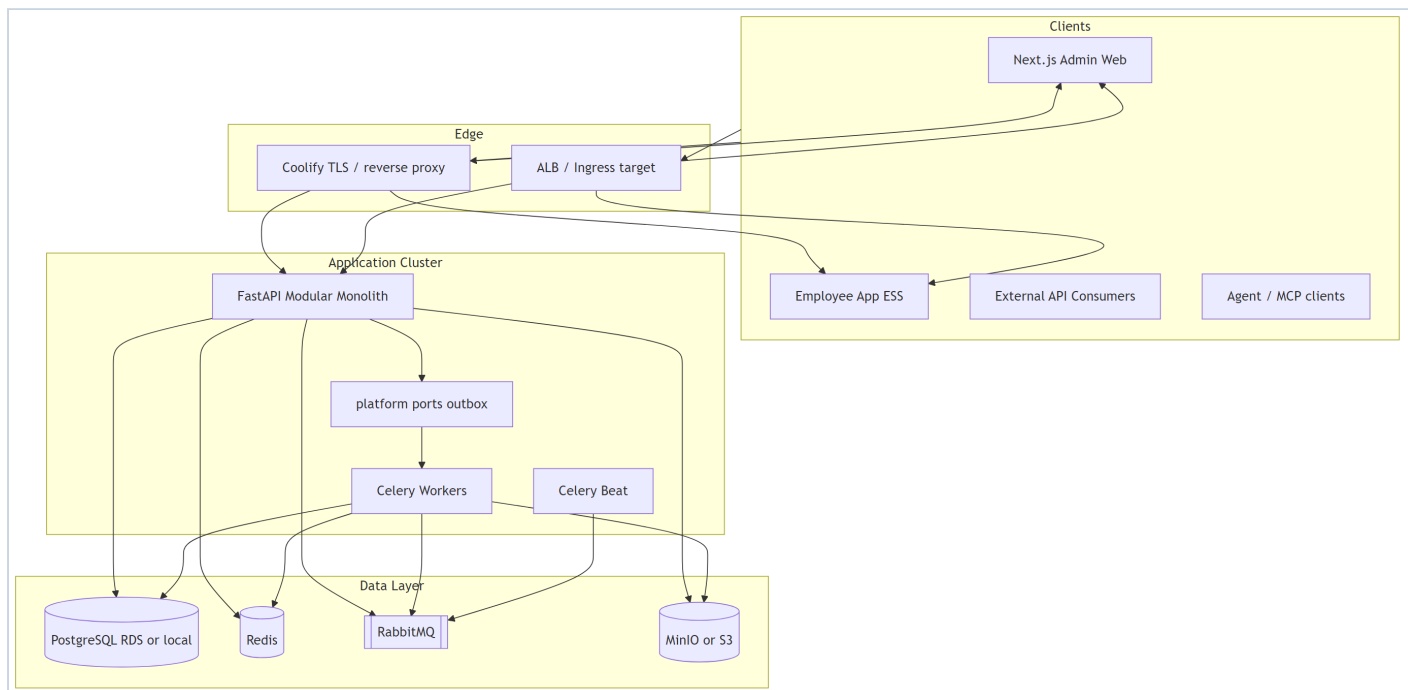
3. Architectural Principles

Principle	Meaning
Modular monolith first	One deployable API; strict module packages; extract services only with evidence
Clean Architecture	Router → Service → Repository → DB; domain free of ORM
DDD bounded contexts	No cross-module table access; integrate via APIs / events / engines
API-first	Versioned REST/OpenAPI before UI
Multi-tenant by construction	<code>tenant_id</code> (and company/branch) on transactional data
Soft delete + audit	No physical DELETE on business tables; audit columns mandatory
Config over customization	Workflows, settings, custom fields — not forks
One artifact, many destinations	Same images for AWS, private cloud, and on-prem
Engines mandatory	Workflow, Notification, Audit, Integration must not be bypassed

Forbidden without EARB/ADR: NestJS, Prisma, MongoDB as primary OLTP, business logic in routers, UI→DB access, cross-module SQL, manual production schema changes.

4. High-Level System Architecture

Figure 2



Layer responsibilities

Layer	Responsibility
Clients	Admin web, employee ESS app, integrators
Edge	TLS termination, WAF, rate limiting, routing
API	AuthN/Z, tenancy middleware, domain routers/services
Workers	Email delivery, async jobs, scheduled syncs
Data	OLTP, cache/sessions, broker, object storage (OpenSearch gated)
Platform	Ports/adapters, transactional outbox, numbering, approvals, encryption, SLO

5. Technology Stack (Current / Locked)

Aligned with ERP Architecture Lock Report v1.1 and Addendum v1.2.

5.1 Stack at a glance

Layer	Primary choice	Role
Frontend (admin)	Next.js 16+ · React · TypeScript · Tailwind · ShadCN · Zod	ERP admin shell and module UIs
Frontend (ESS)	Next.js employee-app	Employee self-service
Backend API	Python 3.13+ · FastAPI · Uvicorn	HTTP API modular monolith
Prod process mgr	Gunicorn + Uvicorn workers	Production API serving
ORM / migrations	SQLAlchemy 2.0 · Alembic	Infrastructure layer only
Validation	Pydantic v2	Request/response schemas
Async jobs	Celery · Celery Beat · RabbitMQ	Notifications, scheduled work
Cache / sessions	Redis	Cache, RBAC cache, sessions, Celery backend
OLTP	PostgreSQL	Single transactional standard
Search	OpenSearch (gated)	Full-text / analytics indexing — not required for current OLTP
Object storage	MinIO (on-prem/local) · AWS S3 (cloud)	Documents, exports, backups
Auth federation	Microsoft Entra ID (OIDC) · JWT HttpOnly cookies	SSO + local auth / MFA
Email	Microsoft Graph adapter	Outbound notification email
Containers	Docker	api , worker , beat , web , employee-app
Antivirus	ClamAV	Upload scanning
Field encryption	EncryptedText / FIELD_ENCRYPTION_KEYS	Tax / KYC / bank / integration secrets
Orchestration today	Docker Compose · Coolify on EC2	Primary SaaS shipping path
Orchestration target	Kubernetes Ready · Terraform Ready	Tier D / EARB when readiness met
Design system	design-system/enterprise-erp-platform/MASTER.md (+ iconnect-plus/)	Data-dense · Swiss minimal · UI/UX Pro Max

5.2 Repository layout

```
enterprise-erp/ # iConnect Plus monorepo
├─ apps/api/      # FastAPI modular monolith (+ platform, alembic, workers)
├─ apps/web/      # Next.js admin (iconnectplus.in)
├─ apps/employee-app/ # Next.js ESS PWA (ess.iconnectplus.in)
├─ apps/employee-mobile/ # Employee mobile
├─ docs/          # Baseline pack + 00_CURRENT living docs
├─ design-system/  # UI/UX Pro Max tokens
├─ docker-compose.local.yml # All-in-one local
├─ docker-compose.infra.yml # Redis RabbitMQ MinIO (+ optional OpenSearch)
├─ docker-compose.app.yml # App against external infra
├─ docker-compose.coolify.yml # Coolify / EC2 SaaS profile
└─ .env.example
```

5.3 Backend module packages (mounted)

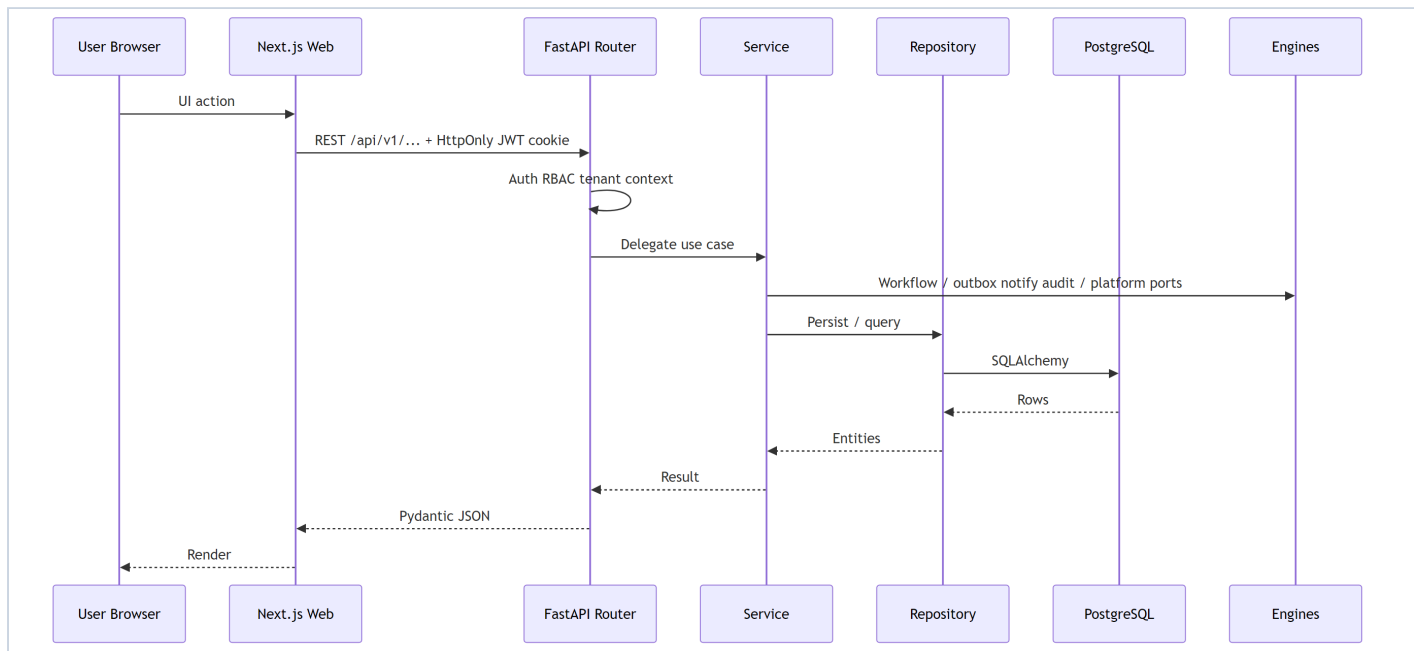
landing, platform, foundation, organization, master_data, finance, sales, procurement, inventory, manufacturing, quality, crm, hr, ess, payroll, recruitment, project, asset, service, helpdesk, document, marketing, grc, analytics, integration, ecommerce, portal, voice_agent, agent_read, mcp_server.

Mandatory flow inside every module:

Router → Service → Repository → Database

6. Clean Architecture and Request Flow

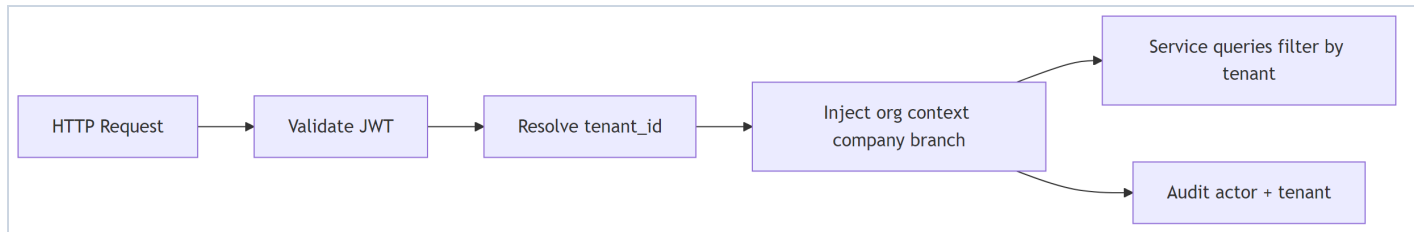
Figure 3



7. Multi-Tenancy and Security

7.1 Tenancy model

Figure 4



Concern	Rule
Isolation	<code>tenant_id</code> on transactional tables; company/branch scoping where applicable
Auth	JWT in HttpOnly cookies; MFA; Microsoft Entra OAuth
AuthZ	RBAC: role → permissions; module membership; org scope
Secrets	Env / secret store — never commit production secrets
Data	Soft delete; audit columns <code>created_at/by</code> , <code>updated_at/by</code> , <code>version</code>
Transport	TLS everywhere in deployed environments

7.2 Security controls (current + target)

Control	Status
HttpOnly session cookies; shorter privileged TTL	Current
RBAC permission catalogs + CI route/UI sync	Current
ClamAV upload scanning	Current
Field-level encryption (tax / KYC / bank / secrets)	Current
SSRF allowlist for integration outbound	Current
Immutable audit via foundation + platform facade/outbox	Current
Edge WAF/DDoS · ISO 27001 / SOC 2 SaaS posture	Target
Security evidence	<code>docs/security/</code>

8. Data and Storage Architecture

8.1 Logical layout

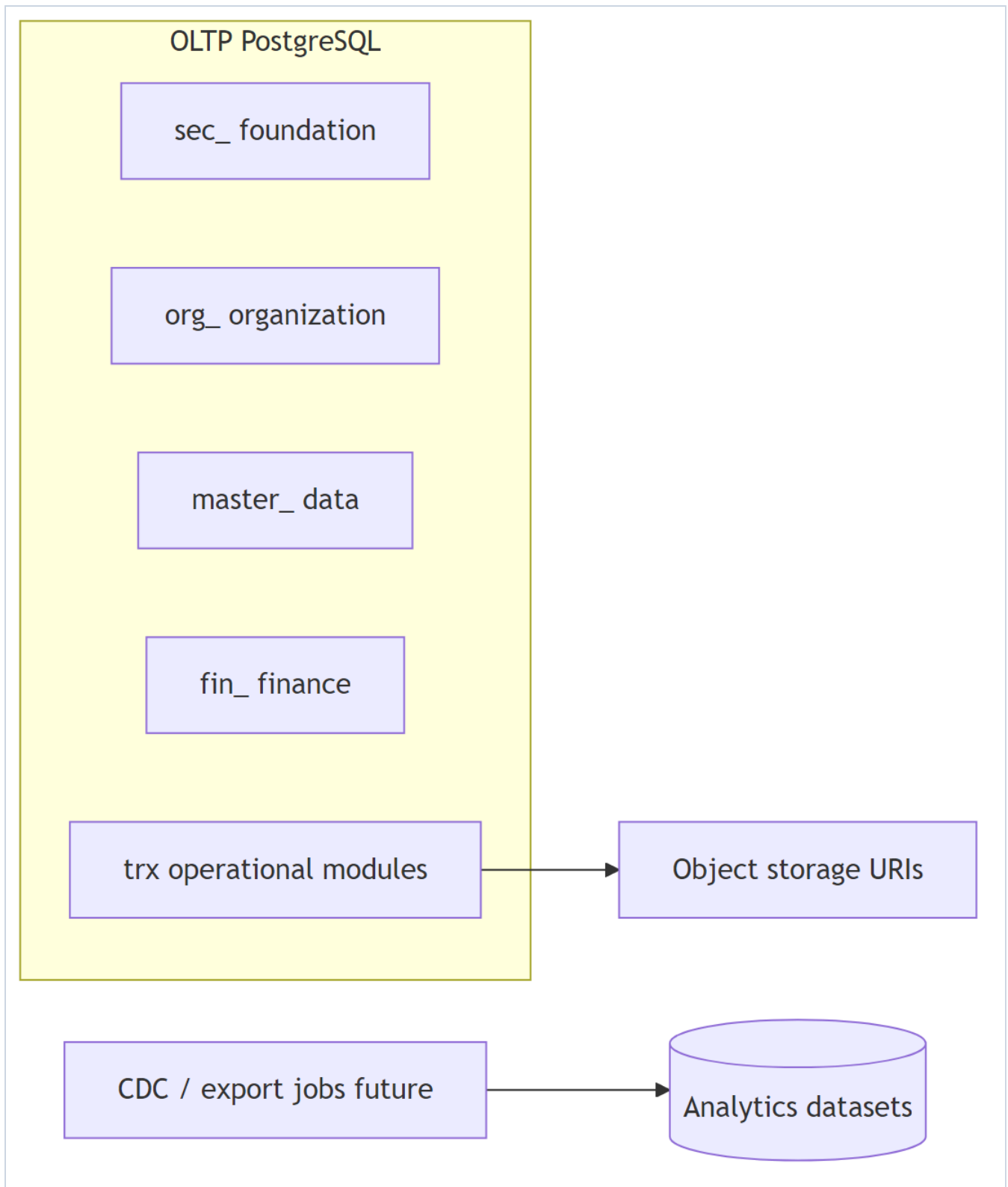
Store	Holds	Isolation
PostgreSQL	All module transactional data, masters, workflow/notification/audit	Tenant (+ company/branch) filters
Redis	Cache, sessions, rate limits, Celery results	Key prefixes per tenant where applicable
RabbitMQ	Task queues	Vhost / queue naming by env
OpenSearch (gated)	Search indexes when enabled	Index or filter per tenant

Store	Holds	Isolation
MinIO / S3	Documents, attachments, exports, backup objects	Prefix/bucket per tenant

8.2 Database standards (DBS)

- UUID primary keys
- Table/prefix standards per domain
- Alembic-only schema changes
- Soft delete — no physical DELETE on business tables
- No cross-module foreign-key coupling across bounded contexts

Figure 5

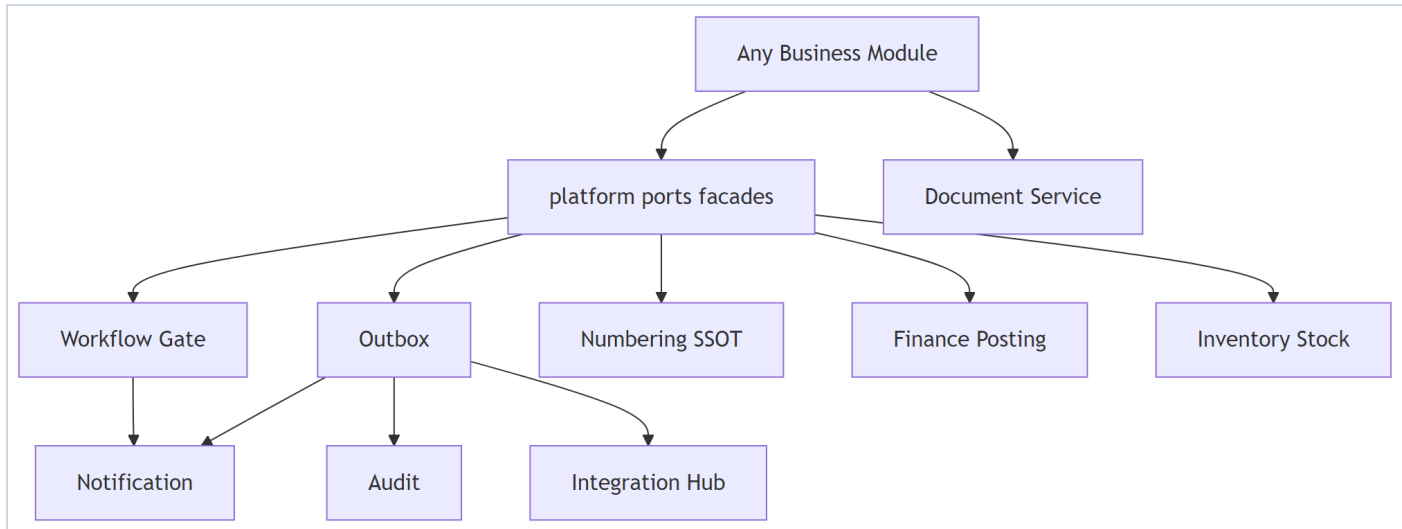


9. Platform Engines

Engine	Responsibility	Home
Identity & RBAC	Login, MFA, OAuth, roles, permissions, module members	Foundation
Workflow	Definitions, instances, approvals, escalation via workflow_gate	Foundation + platform/policies

Engine	Responsibility	Home
Notification / Email	Templates, events, deliveries, Graph send, FCM devices	Foundation + Email UI · PlatformNotifyFacade
Audit	Immutable mutation/event logs	Foundation · PlatformAuditFacade
Transactional outbox	Reliable side effects (notify/audit/finance/integration/domain)	platform/outbox + Celery drain
Numbering SSOT	Document sequences	FndDocumentSequence / platform numbering
Finance posting	Single GL posting path	IFinancePosting adapter
Inventory stock	Single stock mutation path	IInventoryStock adapter
Integration Hub	Connectors, webhooks, sync jobs, mappings, DLQ	Integration module
Document service	Versioned library consumed cross-module	Document module
Settings	System and tenant configuration	Foundation
SLO / metrics	Per-process samples · Prometheus text	/platform/slo/*, /platform/metrics

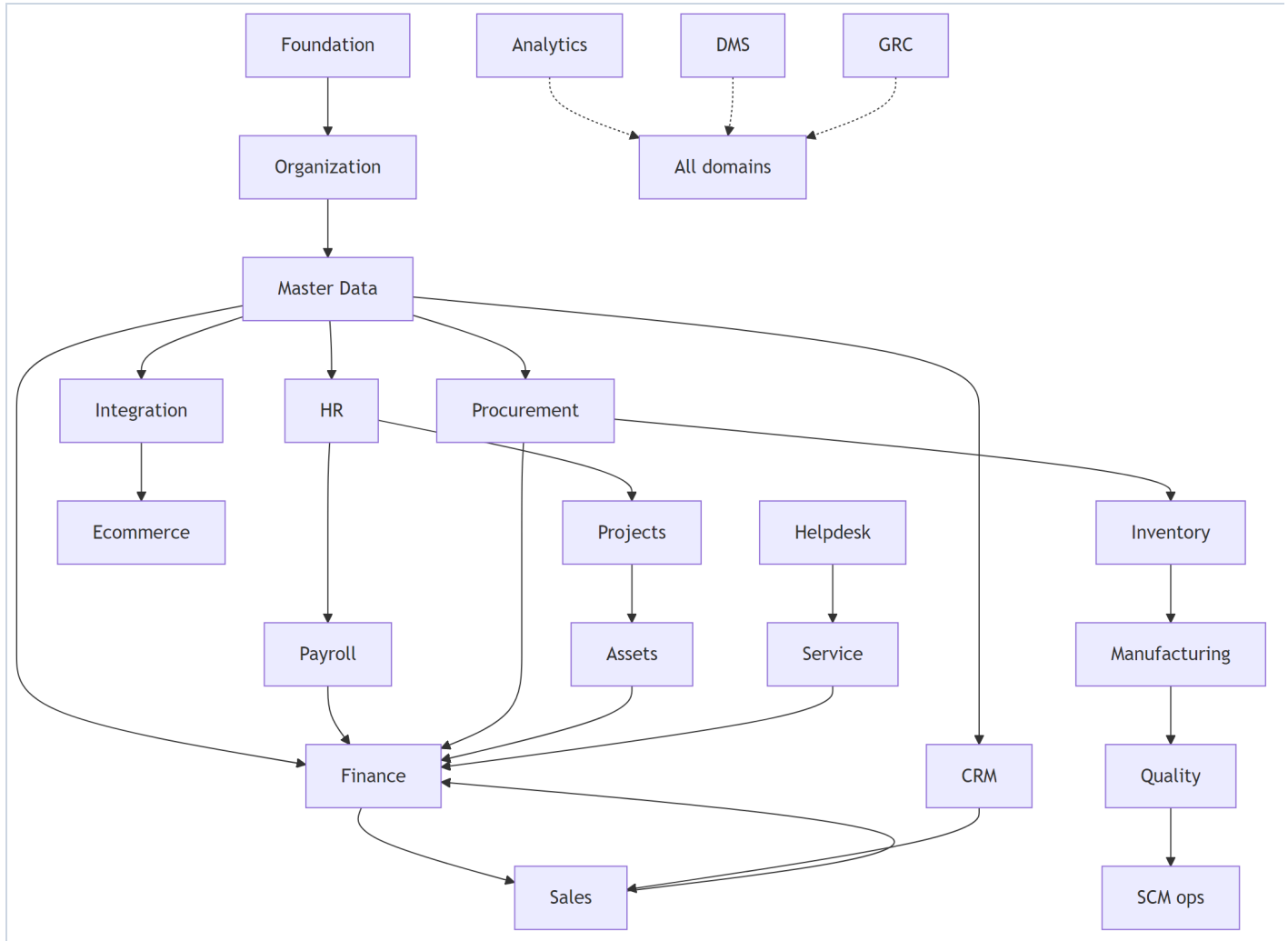
Figure 6



10. Cross-Domain Business Flows

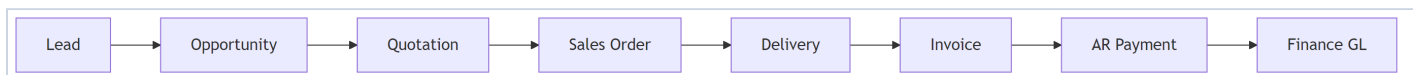
10.1 Enterprise dependency chain

Figure 7



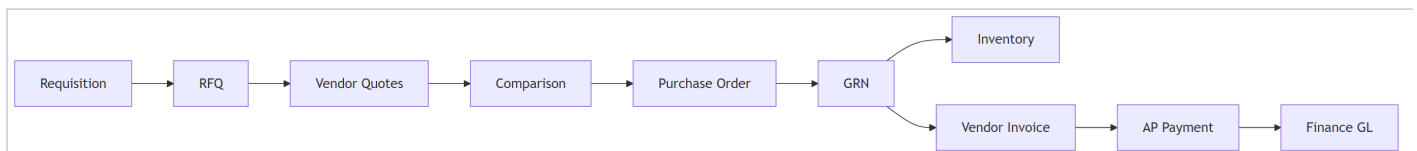
10.2 Order-to-cash

Figure 8



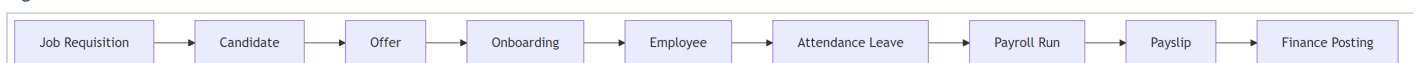
10.3 Procure-to-pay

Figure 9



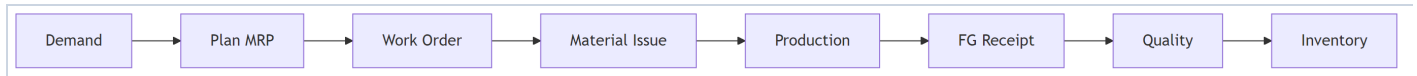
10.4 Hire-to-pay

Figure 10



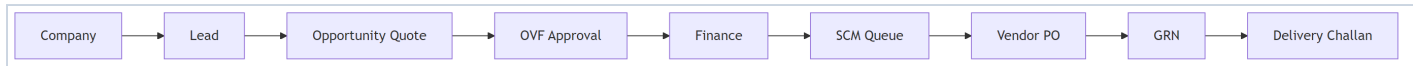
10.5 Plan-to-produce

Figure 11



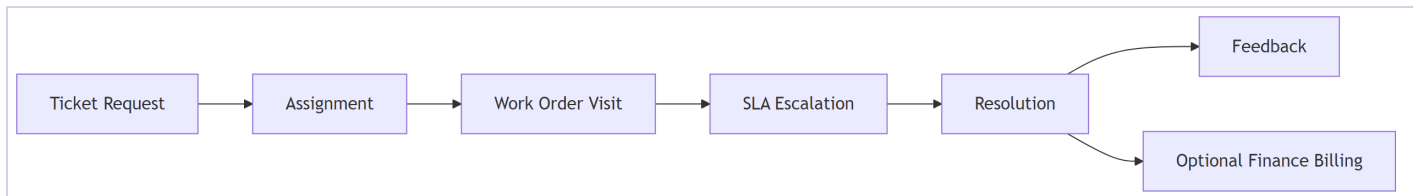
10.6 CRM → SCM bridge (product path)

Figure 12



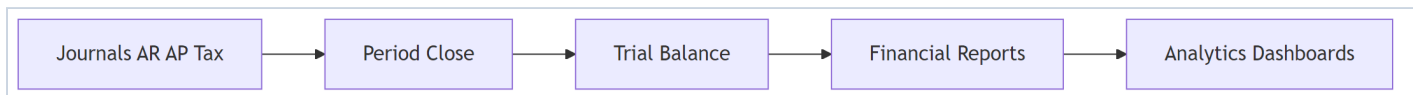
10.7 Service / ITSM

Figure 13



10.8 Record-to-report

Figure 14



11. Module Catalog — Full Portfolio

Maturity legend: **Implemented** · **Partial** · **Scaffold** · **Roadmap**.

11.1 Foundation (Auth, RBAC, Workflow, Audit, Settings)

Web	/foundation
API	/tenants, /users, /roles, /permissions, /workflows/*, /audit/*, /settings, /auth/*
Maturity	Implemented

Features

- Local login, MFA verify, refresh/logout, /me
- Microsoft Entra OAuth (config, login, callback, exchange)
- Tenants, users, roles, permissions
- Module admin/member assignment
- RBAC with Redis-cached permission resolution
- Org context (tenant / company / branch)
- Workflow definitions and instances
- Audit logs
- System settings

Key flows: Authenticate → session → authorized API call; submit → approve/reject/escalate workflow; every mutation audited.

11.2 Email & Notification System

Web	/email
API	/notifications/*, /notifications/email/*
Maturity	Implemented

Features

- Notification templates, events, deliveries
- Microsoft Graph email send, test, overview KPIs
- Compose UI; delivery and event logs
- Device token registration (FCM) for push path
- Celery-backed async delivery

Key flows: Event → template render → Graph/FCM delivery → status logging.

11.3 Organization

Web	/organization
API	/companies, /branches, /departments, /business-units, /locations, /cost-centers, /profit-centers, /organization/tree
Maturity	Implemented

Features: Multi-company hierarchy; branches; departments; BUs; locations; cost/profit centers; org tree.

11.4 Master Data

Web	/master-data
API	/employees, /customers, /vendors, /products, /product-categories, /uoms, /currencies, /taxes, /assets, /warehouses
Maturity	Implemented

Features: Central masters consumed by all operational modules; uniqueness and governance; no duplicate masters inside business modules.

11.5 Finance & Accounting

Web	/finance/** (custom UX)
API	/finance/*
Maturity	Implemented

Features

- Account groups & chart of accounts
- Fiscal years and periods
- Journals (lines, submit, approve, reject, post, reverse)
- GL inquiry, trial balance preview, account ledger
- AR / AP workspaces and aging
- Tax register, currency rates, asset transactions
- Reports: TB, BS, P&L, cash flow, journal register, tax summary, cost center, aging
- Posting / journal / ledger / balance / period-closing engines
- Finance scope / SoD governance hooks

Key flows: Draft journal → approve → post → GL; subledger activity → financial statements; period close.

11.6 CRM

Web	/crm
API	/crm/*
Maturity	Implemented

Features: Companies, leads, assignments, activities, opportunities, stages, pipelines, quotes, OVF, contacts, campaigns, interactions, tasks, follow-ups, meetings, call/email/visit logs, CSAT/feedback.

Key flows: Company → Lead → Opportunity → Quote → OVF → Won/Lost; campaign membership.

11.7 Sales

Web	/sales
API	/sales/*
Maturity	Implemented

Features: Price lists, discount rules, customer credit, quotations, sales orders, deliveries, invoices, returns.

Key flows: Quotation → Order → Delivery → Invoice → Payment/AR; credit holds; returns/credit notes.

11.8 Procurement & SCM Operations

Web	/procurement
API	/procurement/*
Maturity	Implemented (advanced SCM planning Partial vs FRD-15)

Features: SCM queue, requisitions, RFQs, vendor quotations, comparisons, POs, GRNs, delivery challan/status, vendor invoices, returns, contracts, vendor performance.

Key flows: P2P full cycle; OVF → SCM queue → PO → GRN → challan.

11.9 Inventory & Warehouse

Web	/inventory
API	/inventory/*
Maturity	Implemented

Features: Stock, bins, batches, serials, reservations, transfers, adjustments, cycle counts, policies, valuation, reports.

Key flows: GRN putaway → reserve → issue/transfer → count/adjust → valuation.

11.10 Manufacturing

Web	/manufacturing
API	/manufacturing/*
Maturity	Implemented

Features: BOMs, routings, work centers, machines, production orders, material issues/returns, production receipts, scrap, WIP, variances.

Key flows: Plan → WO → issue → execute → FG receipt → WIP/variance close.

11.11 Quality Management

Web	/quality
API	/quality/*
Maturity	Implemented

Features: Inspection/sampling plans, characteristics, defect types, incoming/in-process/final inspections, NCRs, CAPAs, supplier quality, complaints, audits, scores.

Key flows: Plan → inspect → release/hold; NCR → CAPA.

11.12 HRMS

Web	/hr
API	/hr/*
Maturity	Implemented

Features: Designations, profiles, employment, shifts, holidays, leave types/balances/requests, attendance, documents, performance, goals, appraisals, training, separation.

Key flows: Hire-to-retire; leave approval; attendance → payroll handoff.

11.13 Recruitment

Web	/recruitment
API	/recruitment/*
Maturity	Implemented

Features: Requisitions, postings, sources, recruiters, candidates, applications/stages, interviews/feedback, offers, BGV, talent pools, onboarding tasks.

Key flows: Requisition → apply → interview → offer → BGV → onboard → HR employee.

11.14 Employee Self-Service (ESS)

Web	Employee app (not admin registry)
API	/ess/*
Maturity	Implemented (API + employee-app)

Features: Profile, leave, attendance corrections, on-duty/comp-off, payslips, KYC/bank, documents, assets, performance, holidays, announcements, device tokens, notifications.

11.15 Payroll

Web	/payroll
API	/payroll/*
Maturity	Implemented

Features: Periods, salary structures/components, employee salaries, earning/deduction types, tax/statutory, runs, payslips, bonuses, reimbursements, loans, adjustments, summaries, Finance posting.

Key flows: Attendance/leave → run → approve → payslip → GL → bank file (FRD).

11.16 Project Management

Web	/projects
API	/projects/*
Maturity	Implemented

Features: Projects, phases, milestones, tasks, timesheets, resource plans/allocations, budgets, costs, issues, risks, change requests, documents.

Key flows: Approve → plan → execute → timesheet cost → close / bill.

11.17 Asset Management

Web	/assets
API	/assets/*
Maturity	Implemented

Features: Categories, assets, components, assignments, transfers, locations, warranties, insurance, maintenance plans/work, depreciation, disposals, audits, meter readings.

Key flows: Register → assign → maintain → depreciate → dispose.

11.18 Service Management

Web	/service
API	/service/*
Maturity	Implemented

Features: Categories, requests, tickets, assignments, schedules, work orders, tasks, visits, materials, time entries, SLAs, escalations, contracts, feedback.

Key flows: Request → assign → WO/visit → complete → feedback → optional billing.

11.19 Helpdesk

Web	/helpdesk
API	/helpdesk/*
Maturity	Implemented

Features: Categories, priorities, tickets, assignments, comments, SLAs, escalations, knowledge base/articles, resolutions, support teams, shifts, schedules, feedback.

Key flows: Report → assign → resolve → CSAT; SLA breach → escalate.

11.20 Document Management (DMS)

Web	/documents
API	/documents/*
Maturity	Partial / scaffold-rich (storage/OCR hardening ongoing)

Features: Folders, documents, versions, tags, permissions, shares, approvals, workflows, templates, retention, archives, comments, checkouts, audits, attachments, reports.

Key flows: Upload → review → approve → publish; retain → archive.

11.21 GRC (Governance, Risk, Compliance)

Web	/grc
API	/grc/*
Maturity	Scaffold / broad CRUD (+ dashboard)

Features: Policies/versions/acknowledgements, controls/tests, risk categories/register/assessments/treatments, compliance frameworks/requirements/assessments, audit plans/audits/findings, corrective actions, exceptions, incidents, notifications, reports.

Key flows: Risk identify → assess → treat → control test; audit → finding → CAPA; policy lifecycle.

11.22 Analytics (BI)

Web	/analytics
API	/analytics/*
Maturity	Implemented

Features: Dashboards, widgets, reports, schedules, datasets, metrics, KPIs, dimensions, alert rules, subscriptions, imports/exports.

Key flows: Dataset → metric/KPI → dashboard/report → schedule/alert.

11.23 Integration Hub

Web	/integration
API	/integration/*
Maturity	Implemented

Features: External systems, connectors, API credentials, OAuth clients, webhooks, event definitions, message/retry/DLQ queues, mappings, sync jobs/logs, rate limits.

Key flows: Connect → map → sync; webhook ingest → retry/DLQ.

11.24 Ecommerce

Web	/ecommerce
API	/ecommerce/*
Maturity	Implemented

Features: Stores, channels, listings, carts, orders, payments, shipments, return requests, coupons, promotions, marketplace connectors.

Key flows: Channel order → pay → fulfill → ERP inventory/sales sync; returns.

11.25 Customer Portal

Web	/portal
API	/portal/*
Maturity	Implemented

Features: Portal accounts, profiles, sessions, dashboards, notifications, messages, order/invoice views, document access, support tickets, service requests, preferences, login audits.

Key flows: Portal login → view commercial docs → raise ticket/service.

11.26 Marketing & Social Media

Web	/marketing
API	/marketing/* + CRM campaigns
Maturity	Implemented (workspace) · Partial vs full FRD depth

Features: Platforms/accounts, content requests, AI/Celery generation pipeline, versions/scores, brand voice, research/trends, competitors, calendar, publish jobs, analytics summary; CRM campaigns; ecommerce promos/coupons adjacent.

Key flows: Campaign → content request → generate → score → calendar → publish/approve.

11.27 Platform AI surfaces (Agent / Voice / MCP)

Web	/voice-agent (+ agent assistant UI)
API	/voice_agent/*, agent_read, mcp_server
Maturity	Partial / in progress

Features: Voice-agent helpers; read-only agent query surface; MCP bootstrap mounted from API `main`. Full LLM gateway / RAG / Virtual E.A. remain roadmap (§12).

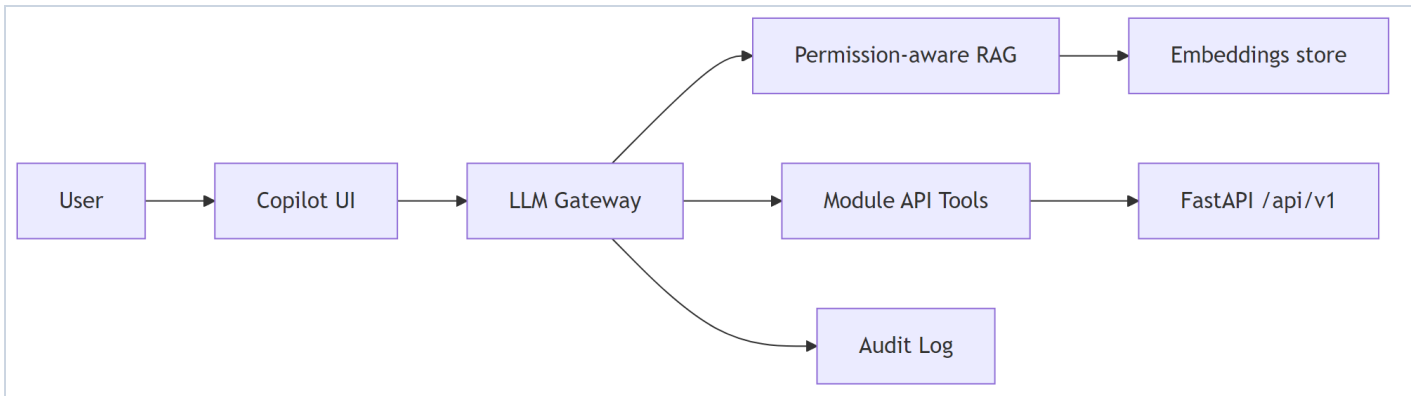
12. Roadmap Platform Extensions

These remain product vision for full GA. Thin API surfaces for agent/voice/MCP already exist (§11.27); Virtual E.A., full Licensing, and Backup & DR product modules are still roadmap.

12.1 AI Assistant & AI Infrastructure

Target features: LLM gateway; permission-aware RAG over ERP + DMS; tool-calling against OpenAPI; in-module copilots; eval/guardrails; audit of AI actions.

Figure 15



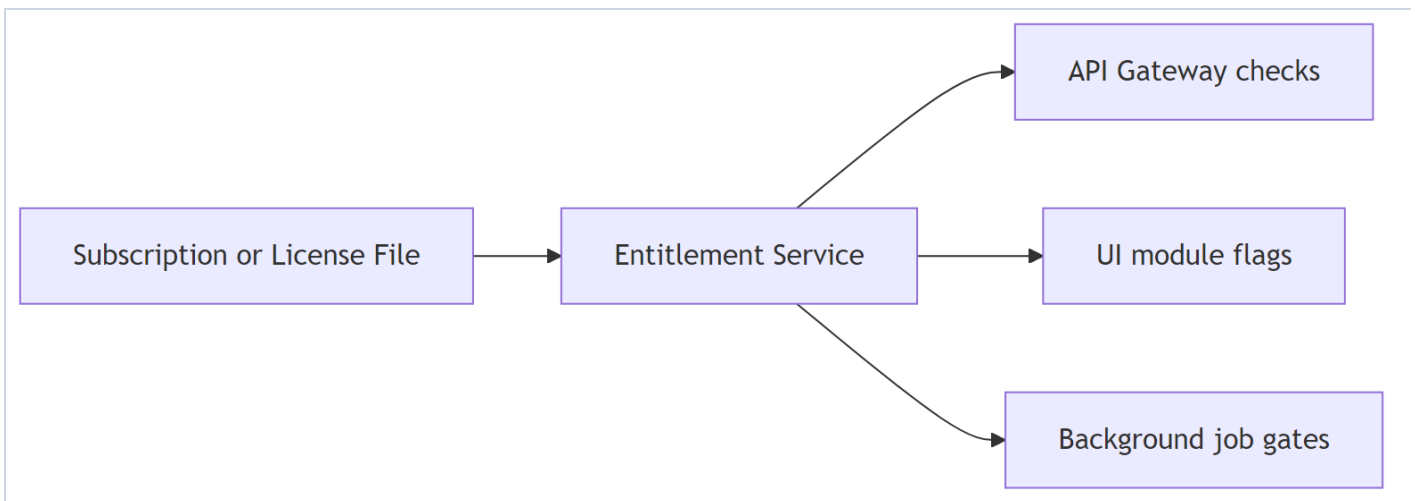
12.2 Virtual Executive Assistant

Target features: Skills/performance graph; assessments/certifications; reminders; calendar/meeting assist; email draft/summarize; proactive insights; human-in-the-loop for consequential actions; consent/transparency controls.

12.3 Licensing & Server-side Activation

Target features: Module entitlements, seats, license keys, online/offline activation, metering, renewal/expiry, grace periods, license audit — required for sell-alone and on-prem.

Figure 16



12.4 Data Backup & DR

Target features: Scheduled full/incremental/WAL backups; encryption; restore verification; per-tenant PITR tooling; DR runbooks; RPO/RTO targets; alignment with GRC BCM.

Ops today: AWS RDS hosting on Coolify profile, `apps/api/scripts/restore_drill.py`, SDD/DBS standards — product Backup & DR module still roadmap.

13. Integration Strategy

13.1 Hub pattern

All external connectivity prefers the Integration Hub: credential vaulting, scheduled/webhook syncs, field mapping, retries, DLQ, connector health.

13.2 Priority connectors

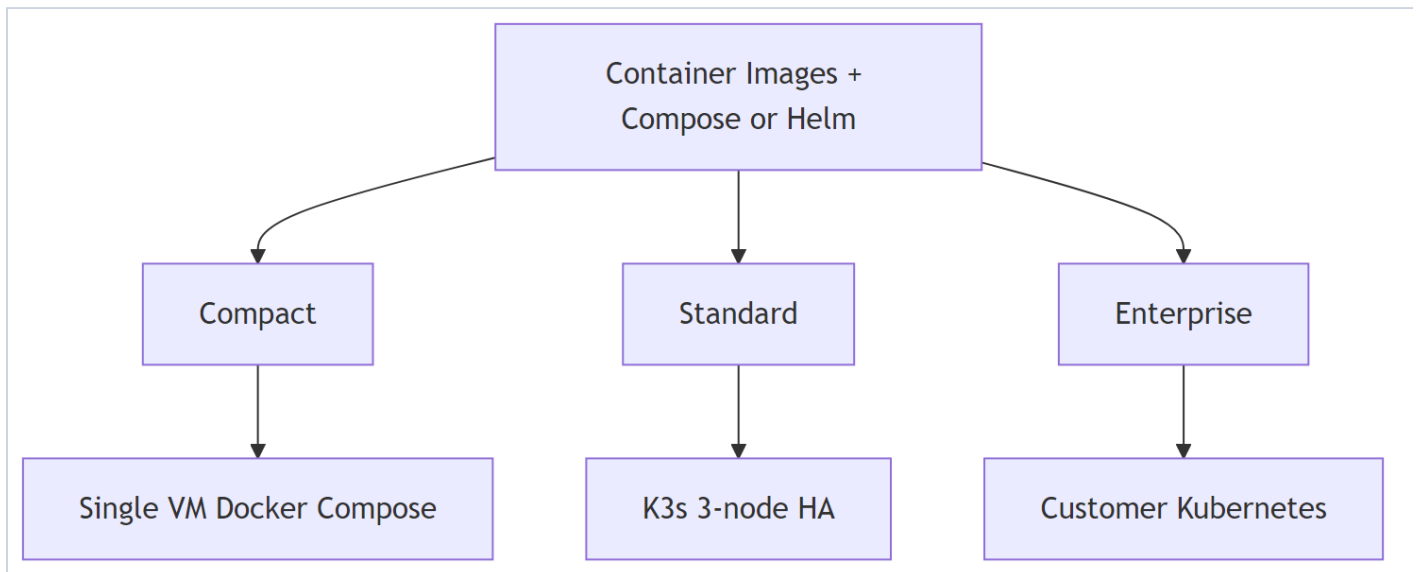
Area	Examples
Identity	Microsoft Entra ID OIDC/SCIM
Email / collab	Microsoft Graph mail
Object storage	MinIO / S3
Payments / banking	Host-to-host, payment status APIs
Tax / e-invoice	Country packs (GST etc.)
Commerce	Shopify/Magento-style channels via Ecommerce module
Signatures	DocuSign / Adobe Sign via DMS
Catch-all	Public OpenAPI, outbound webhooks, CSV/Excel import-export

14. On-Premise Deployment Plan

Same signed container images as cloud; topology changes by customer size.

14.1 Installation profiles

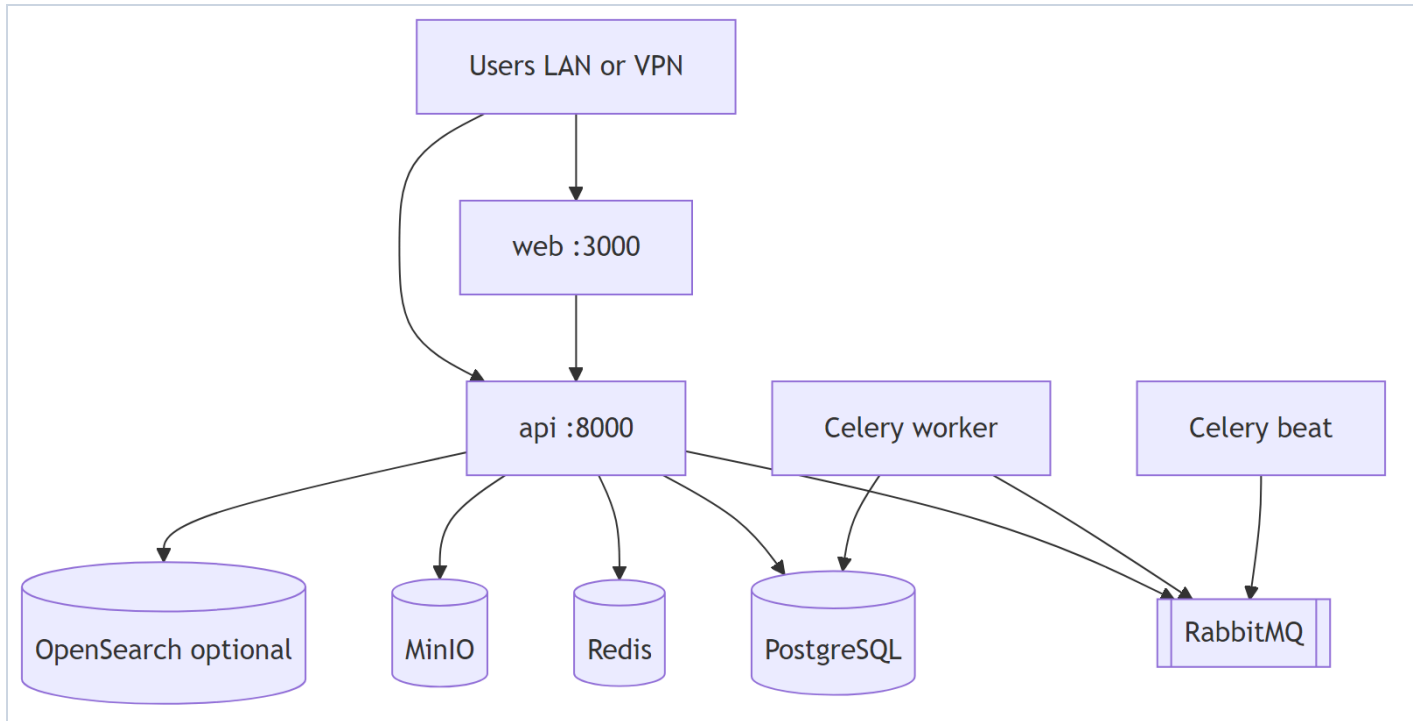
Figure 17



Profile	Target	Shape
Compact	≤ ~100 users	One VM/host: API, worker, beat, web, Postgres, Redis, RabbitMQ, MinIO; optional warm standby
Standard	~100–1,000 users	K3s cluster; HA Postgres (Patroni or managed); replicated Redis/RabbitMQ; MinIO erasure coding
Enterprise	1,000+ / regulated	Customer Kubernetes; customer-managed DB; AD/LDAP federation; air-gap capable; customer DR

14.2 Compact reference topology

Figure 18



14.3 On-prem packaging checklist

Item	Guidance
Installer	CLI (<code>cpctl</code> or equivalent) validates OS, disk, ports; applies Compose/Helm; runs Alembic; health checks
Config	<code>.env</code> / sealed secrets — DB URL, JWT, Graph/SSO optional, storage endpoints
Licensing	Offline signed license files (roadmap entitlement service)
Updates	Online registry pull or offline signed bundle; pre-upgrade backup mandatory
AI	Customer LLM keys, vendor gateway, or local GPU inference — config only, no code fork
Existing artifacts	<code>apps/api/Dockerfile</code> (roles: <code>api</code> / <code>worker</code> / <code>beat</code> / <code>migrate</code>), <code>apps/web/Dockerfile</code> , <code>apps/employee-app/Dockerfile</code> , <code>root docker-compose.yml</code> (infra services)

14.4 On-prem backup & DR

- Nightly DB dump + continuous WAL where feasible
- Object-storage versioning for documents
- Documented restore drill
- Optional secondary site replication for Standard/Enterprise

15. AWS Deployment Plan

15.0 Current SaaS profile (Coolify on EC2) — shipping now

Primary **shipping** path for iConnect Plus SaaS:

Function	Choice
Orchestration	Coolify Compose (<code>docker-compose.coolify.yml</code>) on EC2
Database	Amazon RDS PostgreSQL (<code>DATABASE_URL</code>)
Objects	Amazon S3
Cache / broker	In-compose Redis + RabbitMQ
TLS / DNS	Coolify termination · Hostinger DNS → EC2
Public apps	<code>iconnectplus.in</code> (web) · <code>api.iconnectplus.in</code> · <code>ess.iconnectplus.in</code>

See `docs/09_OPS/coolify-deploy.md`.

15.1 Target cloud mapping (EKS/ECS) — gated

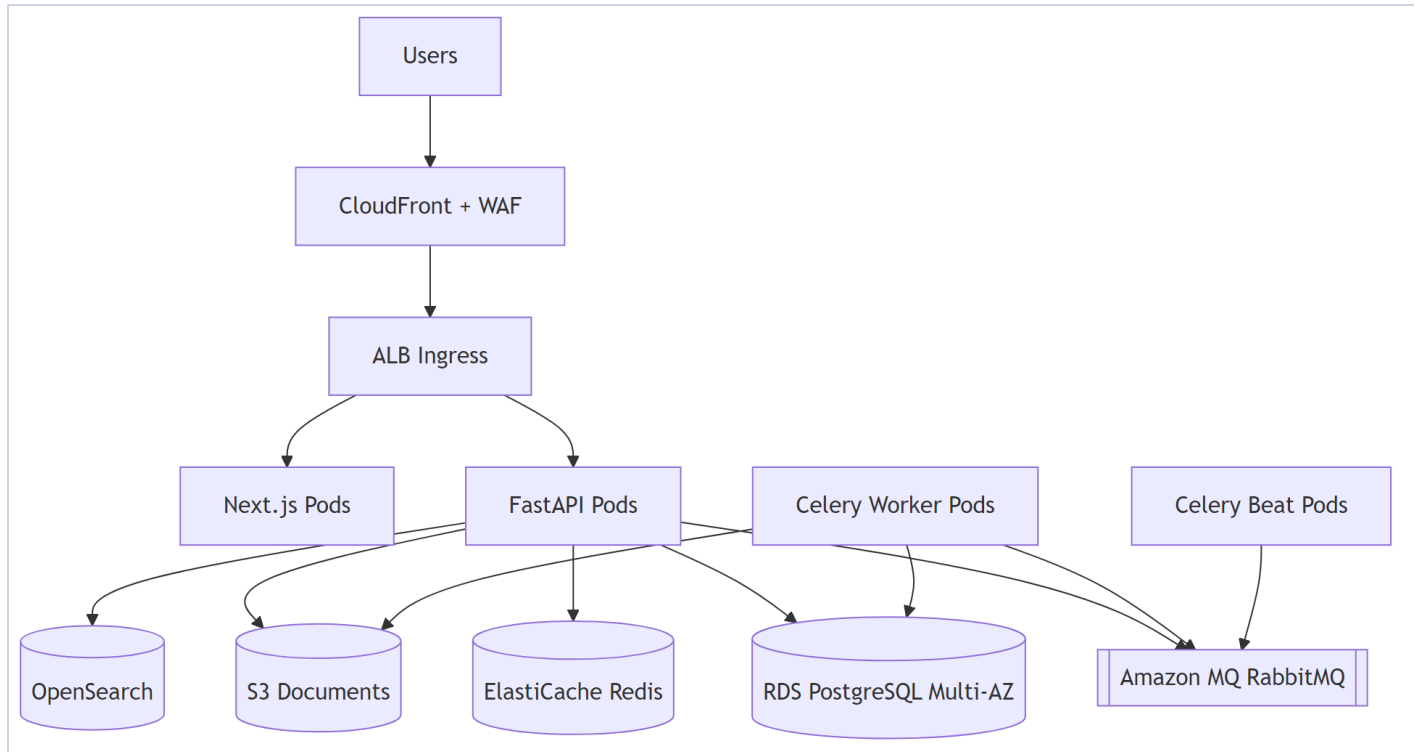
Primary **long-term** cloud target per SDD Volume 4. Do not treat EKS-only as mandatory until `modules/platform/evolution.py` Tier D readiness is true.

15.2 Service mapping

Function	AWS service
Compute	EKS (preferred) or ECS Fargate
API / workers	Kubernetes Deployments or ECS services
Frontend	EKS/ECS + ALB, or CloudFront → S3/SSR containers
Database	Amazon RDS PostgreSQL (Multi-AZ)
Cache	ElastiCache Redis
Broker	Amazon MQ (RabbitMQ) or self-managed on EKS
Search	Amazon OpenSearch Service
Objects	Amazon S3
Secrets	AWS Secrets Manager
TLS / DNS	ACM · Route 53
Edge	CloudFront · AWS WAF
Observability	CloudWatch · OpenTelemetry · Grafana (optional)
IaC	Terraform

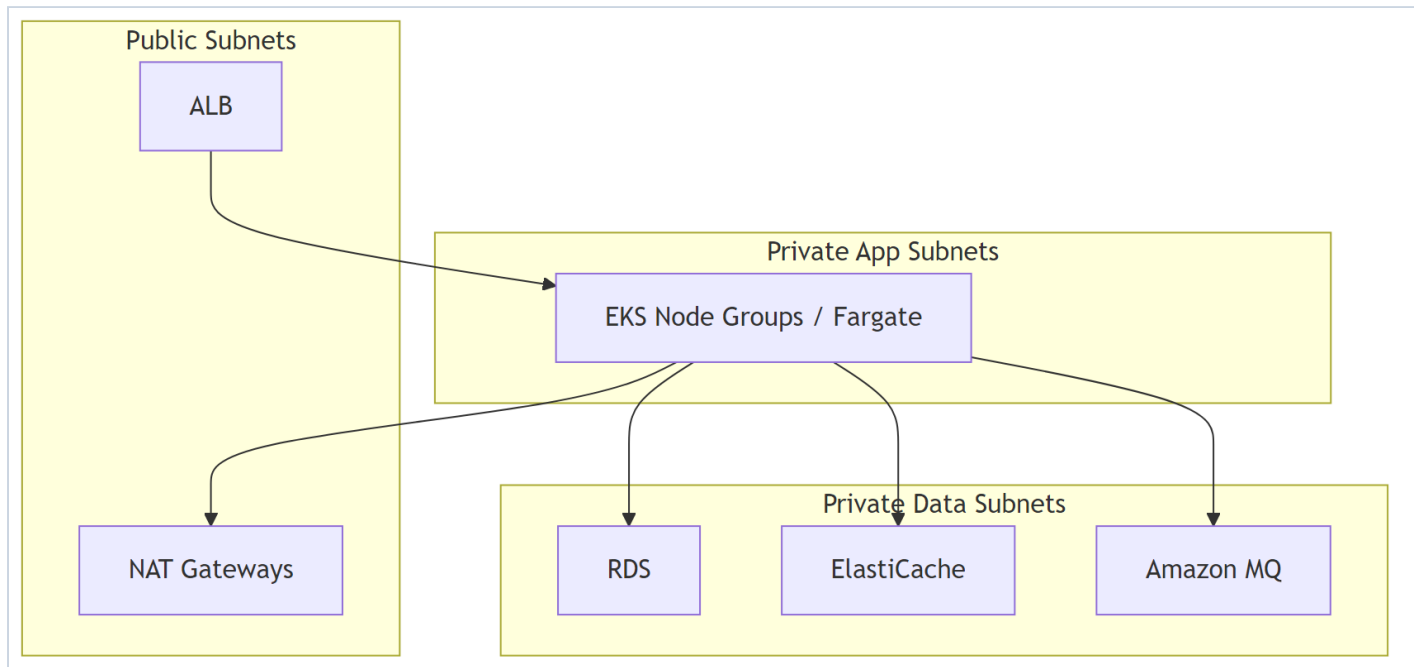
15.3 Target AWS topology

Figure 19



15.4 Network

Figure 20



Rule: Databases and brokers are never publicly accessible.

15.5 Kubernetes workload set

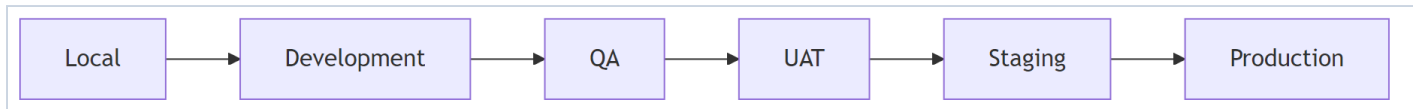
Workload	Image / role	Notes
erp-web	apps/web	HPA on CPU/RPS
erp-api	apps/api role= api	Gunicorn+Uvicorn; HPA

Workload	Image / role	Notes
erp-worker	apps/api role= worker	Scale on queue depth
erp-beat	apps/api role= beat	Single active replica
erp-migrate	Job	Alembic before rollout
erp-employee	apps/employee-app	Optional public path

Ingress, ConfigMaps, Secrets, NetworkPolicies, and PodDisruptionBudgets are mandatory for production.

15.6 Environment promotion (no calendar dates)

Figure 21



Promotion is artifact-based: the **same image digests** move forward; only config/secrets change.

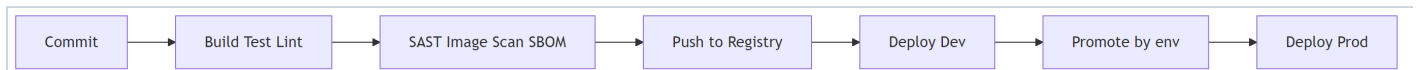
15.7 SaaS control-plane concerns on AWS

- Automated tenant provisioning (schema/DB + seed + DNS)
- Entitlements and metering (Licensing roadmap)
- Per-tenant backup/export
- Status page and error budgets
- Multi-AZ RDS; S3 cross-region replication for DR; defined RPO/RTO

16. DevOps, Environments, and CI/CD

16.1 Target pipeline

Figure 22



Practice	Standard
Source	GitHub
Branches	main, develop, feature/*
CI	Build API/web images; unit/integration tests; architecture/guardrail checks
CD	GitOps (Argo CD) or pipeline deploy to EKS/ECS
Migrations	Forward-only Alembic; expand/contract; run as release Job
Secrets	Secrets Manager / sealed secrets — not git

16.2 Current repo reality

Present	Gap
Dockerfiles for api/web/employee-app	Full-stack prod Compose/Helm not checked in
Infra Compose (Redis, RabbitMQ, MinIO, OpenSearch)	Postgres assumed external / platform-provided

Present	Gap
<code>.env.example</code>	No Terraform/EKS manifests yet
Coolify-style env used operationally	Formalize as one of the deploy profiles

This master document defines the **target**; implementation closes the IaC/CI gaps without changing ADR-001/002.

17. Quality, Observability, and Compliance

Area	Practice
API contracts	OpenAPI; breaking changes fail review
Domain tests	High coverage on finance/payroll math and posting
Module matrix	Standalone degradation tests when licensing lands
E2E golden paths	O2C, P2P, hire-to-pay, record-to-report
Security	Auth/RBAC tests; no cross-tenant leaks
Telemetry	Structured logs, <code>/platform/metrics</code> , <code>/platform/slo/*</code> , traces (OpenTelemetry target)
Architecture CI	Zero cross-module ORM outside adapters; permission catalog sync; OpenAPI uniqueness
Compliance	Audit engine; GRC module; ISO 27001 / SOC 2 target for SaaS

18. Key Risks and Mitigations

Risk	Mitigation
Scope explosion across 20+ modules	Depth before breadth; sellable increments; dependency order Foundation → Org → MDM → business
Tenancy / permission leak	Tenant middleware + query filters; automated cross-tenant tests; pen tests
Payroll / tax correctness	Country packs; domain experts; property-based tests
On-prem support burden	One artifact; LTS channels; installer automation; partner first-line
AI hallucination / unauthorized action	Grounding, citations, user-scoped tools, confirmation, full AI audit
Cursor/AI code drift	<code>.cursor/rules</code> , ADRs in-repo, human review, architecture tests
Backup/DR only manual today	Productize Backup & DR module; expand <code>restore_drill.py</code> ; automate verify restores
Licensing absent	Block sell-alone GA until entitlement service ships
Coolify/EC2 single-node risk	Multi-AZ RDS; backups; plan EKS when Tier D ready

19. Document Hierarchy and Compliance

BRD → FRD (Master + domain FRDs) → SDD v1.1 → DBS v1.1 → ERD → Physical Schema
 → SQLAlchemy Models → Alembic → OpenAPI → Code

Document set	Path
BRD	docs/01_BRD/
FRD	docs/02_FRD/
SDD	docs/03_SDD/
DBS	docs/04_DBS/
Architecture Lock	docs/05_ARCHITECTURE_LOCK/
ERD	docs/06_ERD/
This master	docs/07_MASTER_ARCHITECTURE/
Design system	design-system/enterprise-erp-platform/MASTER.md

No deviation from Architecture Lock v1.1 without EARB approval and an updated ADR.

Appendix A — Connect ownership (implementation focus)

Primary engineer ownership for delivery prioritization:

1. User Login / Access & RBAC
2. Finance
3. AI Assistant & AI Infra
4. Marketing Tool
5. GRC
6. Document Management
7. Virtual E.A.
8. Data Backup & DR
9. Licensing & Server-side Activation
10. Emailer & Notification System

Appendix B — Related source anchors

Concern	Location
API mount list	apps/api/src/shared/router.py
Web module registry	apps/web/src/config/modules.ts
Architecture lock	docs/05_ARCHITECTURE_LOCK/ERP_Architecture_Lock_Report_v1.1.md
Addendum v1.2	docs/05_ARCHITECTURE_LOCK/ERP_Architecture_Addendum_v1.2.md
Living system overview	docs/00_CURRENT/SYSTEM_OVERVIEW.md
Module catalog	docs/00_CURRENT/MODULE_CATALOG.md
Coolify deploy	docs/09_OPS/coolify-deploy.md
Platform package	apps/api/src/modules/platform/
Compose profiles	docker-compose.local.yml · infra · app · coolify
API container entry	apps/api/docker/entrypoint.sh

End of iConnect Plus Master Architecture Document